

Infrastructure Request Report — Database Server & Application Hosting

Department: RWE&Turcas Turkey – Commercial Asset Optimization
Date: March 2026
Classification: Internal – Confidential
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1. Executive Summary

The CAO Team operates an **internal-use-only** energy analytics dashboard that supports daily trading decisions. It is currently hosted on **Render** (a third-party cloud platform) with a managed PostgreSQL database.

The dashboard currently contains no confidential RWE data — only publicly available EPIAS market data and Meteologica weather forecasts. However, hosting on a third-party platform **prevents us from incorporating any confidential or RWE-internal data**, even when there is a clear business need (e.g., proprietary trading models, portfolio positions, internal cost structures). This limitation can only be resolved by moving to RWE-managed infrastructure.

We are requesting **two separate resources**, prioritized by urgency:

Priority	Request	Summary
Project 1 (Primary)	PostgreSQL Database Server	RWE-managed database to store energy market data and enable future confidential data integration.
Project 2 (Secondary)	Application Server	Linux server to host the dashboard web application within the RWE environment.

2. What Does the Dashboard Do?

The dashboard is a web-based analytics platform accessed exclusively by authorized RWE personnel. Key capabilities:

Module	Description
Real-Time Monitoring	DPP and real-time generation tracking for Turkish power plants
DPP Heatmaps	Hourly heatmaps for Natural Gas, Import Coal, Hydro, and Lignite plants
Generation & Demand Analysis	Comparative analysis across fuel types and demand patterns
System Direction Forecasting	ML-based forecasting (CatBoost, LightGBM, XGBoost, Prophet) for YAL–YAT

Module	Description
Forecast Performance	Accuracy tracking of Meteologica generation forecasts
Merit Order Analysis	Forecasted vs. actual merit order comparison
Automated Reporting	Daily heatmap email reports to the trading team

3. Current Setup & Why We Need to Migrate

Current hosting: Render Web Service + Render PostgreSQL + Supabase PostgreSQL (secondary)

All data is currently public (EPIAS + Meteologica). We cannot place confidential data on Render under RWE's data governance policies, which limits the dashboard's analytical potential.

Key reasons for migration:

- **Enable confidential data use** — proprietary analytics, portfolio data, and internal cost models cannot be stored on third-party platforms
- **Corporate governance** — business-critical applications should reside within RWE-managed environments
- **Eliminate vendor dependency** — remove reliance on Render and Supabase
- **Audit & compliance** — better access logging and security controls

4. Technology Stack

Backend: Python 3.11, Flask, Gunicorn, SQLAlchemy, APScheduler, Pandas, NumPy

ML/Forecasting: Darts, Prophet, CatBoost, LightGBM, XGBoost

Visualization: Plotly, Kaleido, XlsxWriter

Frontend: Bootstrap 5, jQuery, Plotly.js

Database: PostgreSQL (with psycopg2 adapter, Alembic migrations)

Backend (Python 3.11)

Library / Framework	Version	Purpose
Flask	2.2.0	Web application framework
Gunicorn	21.2.0	WSGI HTTP server for production deployment
SQLAlchemy	≥ 2.0.0	ORM and database toolkit
Flask-SQLAlchemy	≥ 3.0.0	Flask integration for SQLAlchemy
Flask-Migrate / Alembic	≥ 4.0.0 / ≥ 1.12.0	Database schema migrations
APScheduler	≥ 3.10.4	Background task scheduling (data collection, email reports)
Pandas	≥ 2.0.0	Data manipulation and analysis

Library / Framework	Version	Purpose
NumPy	≥ 1.16.0, < 2.0	Numerical computing
Requests	≥ 2.28.1	HTTP client for external API communication
Flask-CORS	≥ 4.0.0	Cross-Origin Resource Sharing
Flask-Session	0.8.0	Server-side session management
psycopg2	≥ 2.9.9	PostgreSQL database adapter
python-dotenv	1.0.1	Environment variable management

Machine Learning & Forecasting

Library	Version	Purpose
Darts	≥ 0.30.0	Time series forecasting framework
Prophet	1.1.5	Facebook's time series forecasting
CatBoost	≥ 1.1.1	Gradient boosting for time series
LightGBM	4.4.0	Gradient boosting for time series
XGBoost	≥ 2.1.4	Gradient boosting for time series
cmdstanpy	1.2.2	Backend for Prophet

Visualization & Reporting

Library	Version	Purpose
Plotly	5.19.0	Interactive chart generation
Kaleido	0.2.1	Static image export from Plotly (for email reports)
XlsxWriter	≥ 3.1.0	Excel file generation for data exports
openpyxl	≥ 3.1.2	Excel file reading/writing

Frontend

Technology	Purpose
Bootstrap 5.1.3	UI framework
jQuery 3.6.0	DOM manipulation and AJAX
Plotly.js 2.29.1	Interactive client-side charting
Font Awesome 5.15.4	Icon library
ES6 Modules	Modular JavaScript architecture

4.2 External API Dependencies (Outbound Network Access Required)

The application requires **outbound HTTPS access** to the following external services:

Service	URL	Purpose
EPIAS Transparency Platform	https://seffaflik.epias.com.tr/electricity-service/	Turkish electricity market data (DPP, realtime generation, AIC, prices, demand, order summary, merit order data)
EPIAS Authentication	https://giris.epias.com.tr/cas/v1/tickets	TGT token authentication for EPIAS API access
Supabase Database	aws-0-us-east-2.pooler.supabase.com:5432	Meteologica forecast data and EPIAS historical records (to be migrated)
SMTP / SendGrid	Configurable	Automated email report delivery

Note: If the database is migrated to RWE infrastructure, the Supabase dependency can be **fully eliminated** by consolidating all data into the RWE-hosted PostgreSQL instance. The EPIAS API access is mandatory as it is the official source of Turkish electricity market transparency data.

PROJECT 1 (Primary): PostgreSQL Database Server

Most urgent request. Having an RWE-managed database allows us to immediately migrate data away from third-party services and unlocks confidential data integration — even before the application server is migrated.

Requirements

Parameter	Requirement
Engine	PostgreSQL 15+
Storage	50 GB (current data ~1–2 GB; headroom for 10+ years of growth and future confidential data)
Concurrent Connections	20–50
Schemas	Multiple schema support (public , meteologica , epias , and future schemas)

Parameter	Requirement
Admin Privileges	Full DDL/DML — ability to create schemas, tables, indexes, and manage roles
Encryption	SSL/TLS enforced for all connections
Backup	Automated daily backups with point-in-time recovery
Availability	High availability preferred — supports daily trading operations

External / Remote Access (Critical)

The database **must be accessible from outside the RWE corporate network**:

- Before the app server is migrated, the current Render-hosted application needs to connect to it
- Team members need direct access for ad-hoc queries and maintenance (home office, travel)
- Future integration with Databricks / Azure services may require cloud connectivity

Recommended: Secure external endpoint with IP allowlisting + SSL/TLS, or VPN-based access.

Project 1 — Requested Resources

Resource	Specification
1× PostgreSQL Database	Version 15+, 50 GB storage, SSL/TLS, multi-schema support
External Accessibility	VPN, IP allowlisting, or secure external endpoint
Admin Privileges	Full schema/role management capabilities
Backup & Recovery	Automated daily backups

PROJECT 2 (Secondary): Application Server

Secondary request. Once the database is provisioned, deploy the dashboard application on RWE infrastructure to replace Render hosting entirely.

Requirements

Parameter	Requirement
Operating System	Linux (Ubuntu 22.04 LTS or similar)
Python	3.11.x
CPU	8 vCPUs (ML model training is CPU-intensive)
RAM	16 GB (ML models + data processing + Gunicorn workers)
Disk	100 GB SSD (code, dependencies, ML artifacts, logs, future expansion)
Network (Outbound)	HTTPS to EPIAS APIs, SMTP for email
Network (Inbound)	HTTP/HTTPS for dashboard access by RWE personnel

The server also runs **scheduled background tasks** (Europe/Istanbul timezone): daily DPP data updates (16:05), hourly current-version updates, realtime generation updates (05:00 & 12:00), and daily email reports (16:10).

Project 2 — Requested Resources

Resource	Specification
1× Linux Server	8 vCPUs, 16 GB RAM, 100 GB SSD, Python 3.11
Network (Outbound)	HTTPS to seffaflik.epias.com.tr , giris.epias.com.tr , SMTP
Network (Inbound)	HTTP/HTTPS for authorized dashboard users
Database Connectivity	Must reach the PostgreSQL database from Project 1

5. Conclusion

Project 1 (Database) is the primary and most urgent need. The dashboard currently holds only public EPIAS data because we **cannot** store confidential data on third-party infrastructure. An RWE-managed PostgreSQL database would remove this barrier and serve as the data foundation for all current and future CAO analytics.

Project 2 (Application Server) would complete the migration by bringing the entire stack under RWE governance.

Both requests involve modest infrastructure requirements. We are ready to execute upon approval.

For questions or further technical details, please contact the RWE&Turcas CAO Team.